TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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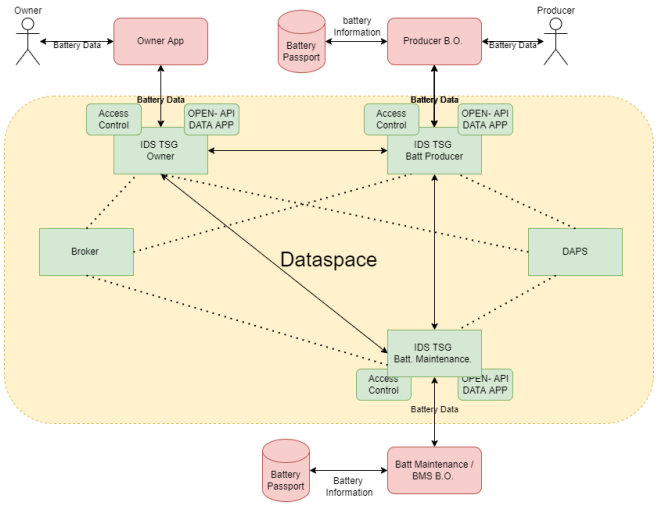
2. Technology Overview

Title of Proposed Technology	Development of an AI diagnosis-based smart battery passport platform for in-use phase under a secure data infrastructure			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-generation Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity (√), AI (√), Next-generation Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification 1)	Technology Field 1 (Large Category)		Technology Field 2 (Large Category)	
	Electrical & Electronics		Information & Communications	
	Medium Category	Medium Category	Medium Category	Small Category
	Batteries	Secondary Batteries	Software	System Integration
Concept of Proposed Technology	The proposed technology aims to develop a dynamic, AI diagnosis-based Smart Battery Passport platform focused on the battery in-use phase, combining physics-informed AI diagnostics with a secure, data-sovereign infrastructure. The platform addresses the growing need for trusted, auditable, and regulation-ready battery lifecycle information, particularly for safety monitoring, residual value assessment, and end-of-life decision support in the European			

market [2].

The core concept integrates a Korean SME's physics-informed AI diagnostic technology with TNO's secure data-sharing infrastructure based on International Data Spaces (IDS), implemented via the TNO Security Gateway (TSG) [1,3,6]. Operational and BMS data from electric vehicles are ingested through an IDS-compliant connector installed at the data owner's boundary, ensuring that data are exchanged without centralization and remain under explicit owner control [3,6].



Within this secure framework, the AI diagnostics continuously generate high-accuracy battery condition indicators, including State of Certified Energy (SoCE), State of Health (SoH), internal resistance evolution, and Remaining Useful Life (RUL) estimates from in-use operational data [7,9]. These indicators are periodically written as dynamic passport updates, forming an objective and time-stamped record of battery condition over its operational lifetime.

In addition, the platform detects and logs anomalies and critical operational events, including safety-relevant and value-relevant negative events, using interpretable, physics-guided analytics [9]. These events are stored as auditable lifecycle evidence, supporting transparent assessment of battery usage history, safety status, and degradation pathways.

A key innovation is the use of accumulated in-use diagnostics and event history to generate end-of-life valuation outputs, such as grading and decision support for EV-to-ESS reuse versus direct recycling, without requiring intrusive or costly physical testing at end of first life [9]. This enables faster, lower-cost, and more trustworthy second-life and recycling decisions.

All data access and exchange are governed by explicit access-control policies, including role-based and contract-based permissions, with full audit logging to ensure traceability and regulatory compliance across stakeholders (OEMs, vehicle owners, service providers, workshops, and recyclers) [1,6,8]. This preserves data sovereignty while enabling controlled multi-party collaboration.

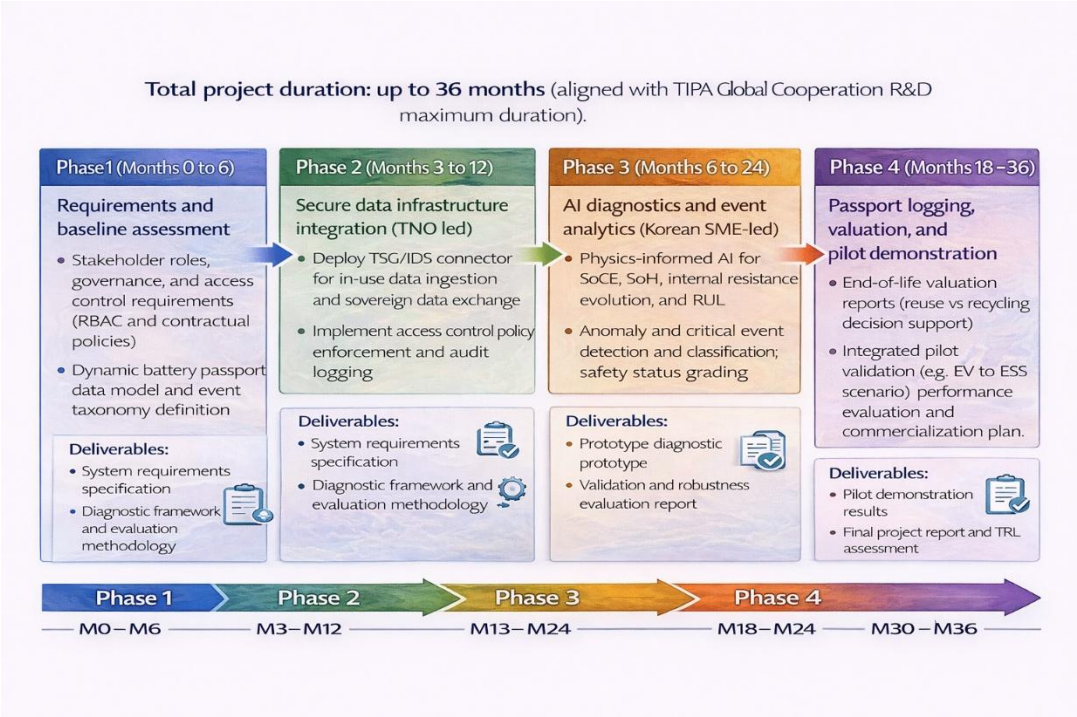
Overall, the proposed technology delivers a trusted Dynamic Battery Passport model for the in-

	use phase, integrating AI-based diagnostics, secure data infrastructure, and lifecycle event logging. It directly supports EU battery regulation compliance, improves confidence in battery lifecycle evidence, and enables scalable deployment toward certified, market-ready battery passport services [2,5,8]
Rationale for Joint R&D Project	Battery lifecycle transparency is becoming a regulatory and market requirement in Europe. In particular, in-use phase information is essential to demonstrate evolving battery performance, safety status, and residual value. Current BMS data can contain measurement errors and may be vulnerable to tampering, which makes it difficult to provide objective, trusted lifecycle evidence for resale, second-life deployment, or recycling decisions. [14] This joint R&D addresses the gap by combining: • TNO: secure data sharing and data sovereignty infrastructure (TSG/IDS) and battery passport architecture alignment • Korean SME: physics-informed AI that produces certified energy and health indicators (SoCE, SoH) and interpretable safety and event analytics The result is a trusted Dynamic Battery Passport model that supports EU market entry and enables transparent certification-oriented evidence from operation through end-of-life. [2]
Key Words	1. Dynamic Battery Passport (in-use phase) 2. SoCE (State of Certified Energy), SoH, RUL, internal resistance 3. IDS (International Data Spaces), TNO Security Gateway (TSG) 4. Data sovereignty, RBAC, access control and audit logging 5. Negative events and critical event logging

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) 2)	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	Deliver an integrated, secure and dynamic battery passport platform for the in-use phase by achieving the following objectives: • Define a joint reference architecture and a dynamic battery passport data model (including event taxonomy) compatible with EU-aligned battery passport initiatives. • Deploy secure in-use data acquisition using TSG (IDS connector) with role-based access control and auditable policy enforcement. • Develop and validate physics-informed AI models to estimate SoCE, SoH, internal resistance evolution and RUL from operational/BMS data, and publish these as passport updates. • Implement anomaly and critical event detection and record negative events and safety status as trusted passport evidence. • Demonstrate end-to-end operation in a pilot workflow (e.g., EV to ESS transition), including end-of-life valuation outputs, targeting TRL 7 or higher.			
R&D Target Specifications	Target specifications (to be finalized during Phase 1 using available pilot datasets and stakeholder requirements): • SoCE and SoH estimation: quantified accuracy and uncertainty reporting against reference measurements. • RUL and internal resistance trend: validated predictive performance over representative duty cycles. • Event detection: defined critical event set, precision/recall targets, and time-to-detect thresholds. • Secure data sharing: demonstrable policy enforcement (RBAC/contract), audit logging completeness, and connector availability/latency. • Passport update performance: near real-time update for key indicators and event logging (latency target agreed with pilot partners).			

R&D Plan



Total project duration: up to 36 months (aligned with TIPA Global Cooperation R&D maximum duration).

Phase 1 (Months 0 to 6): Requirements and baseline assessment

- Stakeholder roles, governance, and access control requirements (RBAC and contractual policies)
- Dynamic battery passport data model and event taxonomy definition

Phase 2 (Months 3 to 12): Secure data infrastructure integration (TNO-led)

- Deploy TSG/IDS connector for in-use data ingestion and sovereign data exchange
- Implement access control policy enforcement and audit logging

Phase 3 (Months 6 to 24): AI diagnostics and event analytics (Korean SME-led)

- Physics-informed AI for SoCE, SoH, internal resistance evolution, and RUL
- Anomaly and critical event detection and classification; safety status grading

Phase 4 (Months 18 to 36): Passport logging, valuation, and pilot demonstration

- End-of-life valuation reports (reuse vs recycling decision support)
- Integrated pilot validation (e.g., EV to ESS scenario), performance evaluation and commercialization plan.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO responsibilities (research institute): • Provide and support deployment of TNO Security Gateway (TSG) as an IDS connector for secure, sovereign in-use data exchange. [9] • Define governance and access control policies (RBAC and contractual rules) and implement audit logging for multi-stakeholder operation. [9] • Ensure battery passport architecture alignment with EU-oriented battery passport frameworks and interoperability requirements (ontology and data model guidance). [9] • Lead integration and contribute to pilot scenario design and validation in Europe-relevant contexts.
Expected R&R of Korean SME Partner	Korean SME's responsibilities: • Provide physics-informed AI diagnostics to estimate SoCE (energy-based health), SoH, internal resistance evolution, and RUL from in-use operational/BMS data. • Implement anomaly and critical event detection and generate trusted safety and negative-event passport entries. • Provide end-of-life valuation logic and reporting for reuse (e.g., ESS) vs recycling decisions using accumulated diagnostics and event history. • Support deployment of AI services in the integrated platform and contribute to pilot data and validation activities.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Prototype of a secure, dynamic Battery Passport service for the in-use phase, integrating: • Sovereign data infrastructure (TSG/IDS) with role-based access control and auditability • Continuous AI-based updates of SoCE, SoH, internal resistance trends and RUL • Trusted logging of negative events and safety-relevant operational history • End-of-life valuation outputs that support EV to ESS reuse decisions and recycling pathways Expected impact: • Higher trust in battery lifecycle evidence (reducing tamper concerns) • Lower cost and higher speed for second-life grading without intrusive testing • Improved EU market readiness and potential contributions to standardization and IP.

6. References

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Appendix A: Key terms (English, from proposal document)

TSG (TNO Security Gateway): IDS-based secure connector software installed at the data owner boundary to enforce permissions and broker data exchange without centralizing data, preserving data sovereignty.

IDS (International Data Spaces): A European-led data sharing architecture enabling secure exchange while maintaining data ownership and control.

SoCE (State of Certified Energy): An energy-based indicator (kWh) reflecting certified remaining capacity, used as a key health-related metric.

Ontology: A shared formal data model defining names, attributes, and relationships for battery passport data to ensure interoperability.

RBAC (Role-Based Access Control): Security mechanism that differentiates read/write permissions by stakeholder role (OEM, owner, workshop, recycler, etc.).

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description
·A consistent and objective indicator of the technological maturity of key element technologies

	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization